PVLoad-R1 Bench Card

ARDUINO → 1×10 HEADER

1	`5V`	5 V
2	`GND`	GND
3	`D6`	K1 drive
4	`D7`	K2 drive
5	`D8`	K3 drive
6	`D9`	CS U2
7	`D10`	CS U1
8	`D11`	SDI
9	`D12`	SDO
10	`D13`	SCK

Header pin 1 is the square pad. On the Uno, `5V` and `GND` are on the power header; `D8`–`D13` sit nearest the USB socket.

BANANA JACKS

J1 red is PV+. **J2** red is +24 V. **J3** black is ground, shared by the supply return and the cell negative — use a stacking plug so nothing is jammed alongside anything else.

POWER UP

1. 24 V on
2. 5 V on (plug in the Arduino)
3. PV− → J3
4. PV+ → J1
5. illuminate the cell

POWER DOWN

1. darken the cell
2. PV+ out of J1
3. PV− out of J3
4. Arduino out (5 V off)
5. 24 V off

Both supply leads stay in J2 and J3 the whole time. **24 V on** and **24 V off** mean the output switch on the supply, not plugging anything in. If your supply has no output switch, seat the black lead in J3 first and treat the red going into J2 as the on step.

CABLES

Arduino. USB-A to USB-B, straight to the laptop.

24 V supply. Two banana leads to J2 and J3. Set 24.0 V, current limit about 100 mA.

EDFA100P. Plain USB cable, no adapter.

The two 617s. They have an IEEE-488 socket and nothing else, so you need a USB-GPIB adapter — National Instruments, to match the VISA on the laptop. One adapter carries both: run a cable from the adapter to the first meter, then a second cable from the first meter to the second, because the connectors stack.

Set a **different address on each meter** from its front panel before connecting them. Both ship on 27 and will collide. Use 22 for the volts meter and 23 for the amps meter.

PUTTING THE TWO 617S IN THE CIRCUIT

Each 617's `INPUT` is a triaxial socket, so you cannot clip a banana lead straight on. Screw a **triax-to-dual-banana adapter** onto it. That gives you two banana posts, **HI** and **LO**.

Amps meter — goes inside the wire
Current can only be measured by making it flow through the meter, so you cut the PV+ line and put the meter in the gap. The wire that used to run from the cell straight to J1 now runs through the meter instead:

```
cell +  →  HI post
LO post  →  J1
```

Volts meter — sits across the cell
Nothing is cut. Two leads simply touch the cell's own two terminals:

```
HI post  →  cell +
LO post  →  cell −
```

Put these on the **cell leads themselves**, not on J1 and J3. On J1 you would be reading through the amps meter as well, which makes the voltage read low.

THE EDFA INTERLOCK

Small connector on the back of the amplifier. It is a laser safety cutout: the amplifier only emits while that circuit is closed, the idea being that you wire it to a door switch so opening the door kills the beam. **Plug in the shorting plug it shipped with.** With nothing in it the laser will not turn on at all, no matter what the software asks for.

LASER — Class 3B, 1550 nm, invisible. It emits up to 30 mW whenever it is enabled, even at zero pump current and with no optical input. Terminate the output before running. Never look into the fibre bulkhead.

WHY THE POWER ORDER MATTERS. With 5 V up, K1 closed and 24 V missing, K1 shorts out R1 and the cell drives straight into the DigiPot's input clamp at 80% of its absolute maximum. That is what kills a chip. The cell goes on last and comes off first for that reason.